



Entergy®

Entergy Operations, Inc.

River Bend Station
5485 U.S. Highway 61N
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Tel 225-381-4157

Steven P. Vercelli

General Manager –Plant Operations

RBG-47762

June 5, 2017

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Licensee Event Report 50-458 / 2017-005-00
River Bend Station – Unit 1
Docket No. 50-458
License No. NPF-47

RB1-17-0062

Dear Sir or Madam:

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Sincerely,

SPV / dhw

Enclosure

cc: U. S. Nuclear Regulatory Commission
Region IV
1600 East Lamar Blvd.
Arlington, TX 76011-4511

NRC Sr. Resident Inspector
P. O. Box 1050
St. Francisville, LA 70775

INPO
(via ICES reporting)

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Central Records Clerk
Public Utility Commission of Texas
1701 N. Congress Ave.
Austin, TX 78711-3326

Department of Environmental Quality
Office of Environmental Compliance
Radiological Emergency Planning and Response Section
Ji Young Wiley
P.O. Box 4312
Baton Rouge, LA 70821-4312

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104			EXPIRES: 03/31/2020	
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block)					Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.				
1. FACILITY NAME River Bend Station – Unit 1					2. DOCKET NUMBER 05000-458			3. PAGE 1 OF 3	
4. TITLE Operations Prohibited by Technical Specifications Due to Inoperable Main Control Room Filter Train									
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME
04	05	2017	2017	005	00	06	05	2017	DOCKET NUMBER 05000
									DOCKET NUMBER 05000
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
1		☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. POWER LEVEL 100		☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)		☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
				☐ 50.73(a)(2)(i)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A	
12. LICENSEE CONTACT FOR THIS LER									
LICENSEE CONTACT Tim Schenk, Manager – Regulatory Assurance							TELEPHONE NUMBER (Include Area Code) 225-381-4177		
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT									
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
none									
14. SUPPLEMENTAL REPORT EXPECTED						15. EXPECTED SUBMISSION DATE		MONTH	DAY
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO									
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On April 5, 2017, during a scheduled surveillance test, it was discovered that the Division 1 main control room ventilation filter train was inoperable due to the flow rate being below the lower limit. The initial investigation found that a manually-operated damper in the flow path for that filter train was not correctly positioned. Work documents from the recent refueling outage that started on January 28 were reviewed, and it was found that the damper had been closed as part of establishing a system configuration needed for repairs to other dampers. It was concluded that the as-left condition of the damper after restoration from the maintenance was such that it vibrated in the closed direction upon the first start of the filter train fan, and that the filter train had thus been inoperable since the return to service on February 26, 2017. The plant was restarted at the end of the refueling outage on March 8. On March 10, an unplanned manual reactor scram was initiated due to a steam leak in the turbine building. The leak was repaired and the plant was restarted on March 11. Since the filter train was inoperable on both occasions, those mode changes were made in violation of Technical Specifications. The investigation of this event also found that during one 8-minute period when the Division 1 filter train was inoperable, the Division 2 main control room fresh air system was inoperable for planned testing. This event was caused by the lack of any instructions provided for the restoration of the Division 1 subsystem that directed the torquing of the damper locking mechanism									

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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River Bend Station – Unit 1	05000-458	YEAR	SEQUENTIAL NUMBER	REV NO.
		2017	005	00

NARRATIVE**REPORTED CONDITION**

On April 5, 2017, during a scheduled surveillance test, it was discovered that the Division 1 main control room ventilation filter train [VI] was inoperable due to the flow rate being below the lower limit. The initial investigation found that a manually-operated damper (**CDMP**) in the flow path for that filter train was not correctly positioned. Work documents from the recent refueling outage that started on January 28 were reviewed, and it was found that the damper had been closed as part of establishing a system configuration needed for repairs to other dampers. It was concluded that the as-left condition of the damper after restoration from the maintenance was such that it vibrated in the closed direction upon the first start of the filter train fan, and that the filter train had thus been inoperable since the return to service on February 26, 2017.

The plant was restarted at the end of the refueling outage on March 8. On March 10, an unplanned manual reactor scram was initiated due to a steam leak in the turbine building. The leak was repaired and the plant was restarted on March 11. Since the filter train was inoperable on both occasions, those mode changes were made in violation of Technical Specifications. These conditions are being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as operations prohibited by Technical Specifications.

The investigation of this event also found that during one 8-minute period when the Division 1 filter train was inoperable, the Division 2 main control room fresh air system was inoperable for planned testing. This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as a potential loss of safety function of the main control ventilation system.

INVESTIGATION

The manually-operated damper was closed as part of establishing a work boundary for maintenance being performed in other sections of the system. The normal position of the damper is partially open, as established when the system was originally flow balanced. The operating mechanism includes a bolted linkage to lock the damper in place. A match mark is on the mechanism indicating the correct position of the damper.

The documentation for the restoration of the system to its operating configuration was reviewed and it was found that there were no instructions regarding the need to torque the bolts in the damper locking mechanism. It was concluded that the damper was re-opened to align with the match mark, but that the locking bolts were not torqued. It could not be determined from the documentation who had re-opened the damper, so the actual as-left condition of the locking bolts could not be determined.

The damper was correctly locked into position on April 6.

The same damper on the Division 2 subsystem was also manipulated during the work, but it has since been verified that the locking bolts for that damper were adequately torqued upon restoration.

CAUSAL ANALYSIS

This condition was caused by deficient work instructions, in that no torque specifications were provided in the work documents for restoring the damper to its normal position.

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CONTINUATION SHEET**

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				2017	005	00

CORRECTIVE ACTION TO PREVENT RECURRENCE

The applicable surveillance test procedures will be revised to include instructions on proper torqueing of the locking bolts of the manually-operated dampers. This action is being tracked in the corrective action program.

PREVIOUS OCCURRENCE EVALUATION

No similar events have been reported by River Bend Station in the last three years.

SAFETY SIGNIFICANCE

During the 8-minute period when both divisions of the main control room fresh air system were inoperable on March 15, the Division 2 subsystem was under the direct control of a dedicated operator as part of planned testing. That subsystem was thus available for service had it been necessary. At all other times, the Division 2 subsystem was operable and able to fulfill its design safety function. This event, therefore, did not constitute an actual loss of safety function. This event was of minimal significance to the health and safety of the public.

(NOTE: Energy Industry Identification System component function identifier and system name of each component or system referred to in the LER are annotated as (**XX**) and [XX], respectively.)